



Introduction to DSP and DSP system

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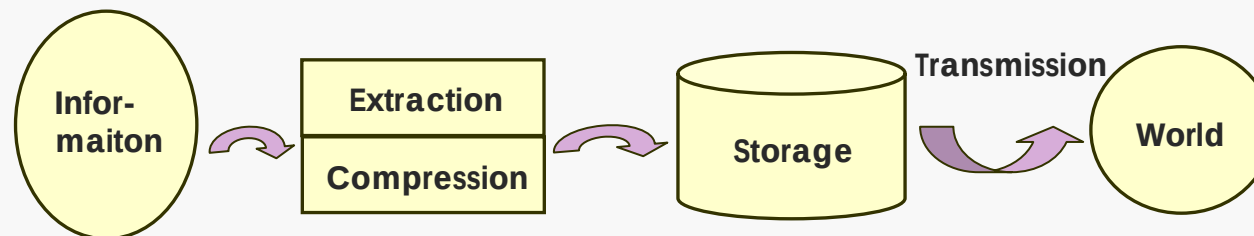
- ◇ Introduction to signal processing concept
- ◇ Benefits of DSP over ASP
- ◇ DSP basic operations
- ◇ Examples of DSP applications
- ◇ Implementation methods of DSP algorithms
- ◇ DSP system design & design phases
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Signal Processing Concept

◇ Signal Processing

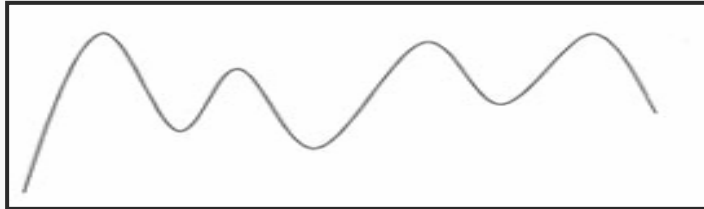
- Fundamental to information processing
- Includes various methods
 - For extracting information from nature or machines
 - For reducing information content in a signal
 - For transforming the signal into a form suitable for transmission or storage
- Mostly needs real-time processing and digital processing



Analog signal

◇ Analog Signal

- Signal in continuous variable-shaped form
- Natural signal format in human environment



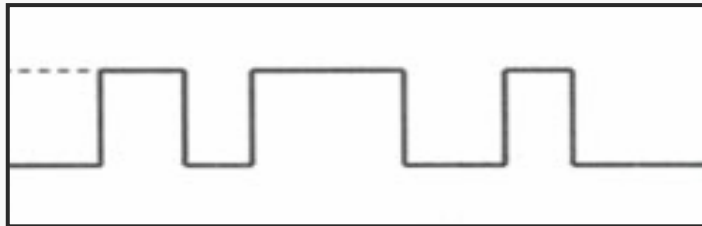
- Advantage
 - Comfortable & natural (no needs for conversion)
 - Compatible with human environment
- Disadvantage
 - Very weak in noise (additive to the signal)
 - Difficult to send over long distance



Digital Signal

◇ Digital Signal

- Signal in a discrete set of binary value (0, 1)
- Used in digital & computer system



- Advantage
 - Very reliable
 - Very strong in noise environment
- Disadvantage
 - Needs A/D, D/A conversion
 - Limited in resolution



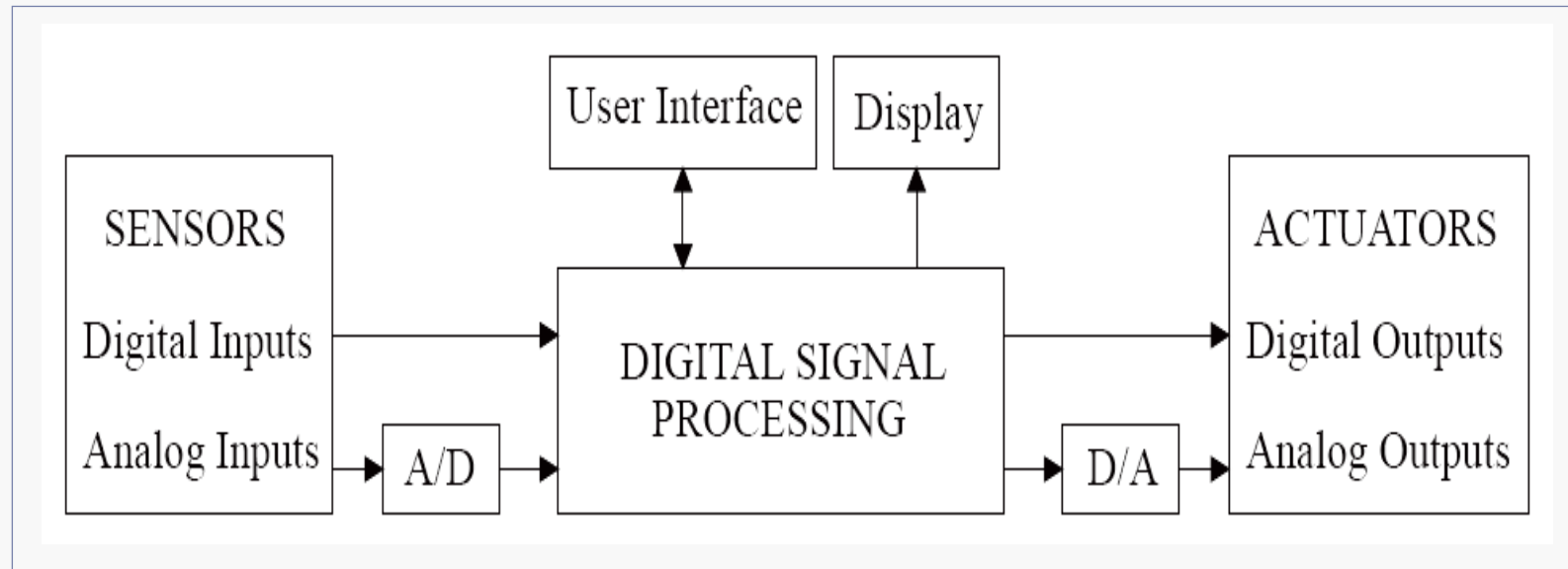
Benefits of DSP over ASP

- ◇ Limitations of analog signal processing
 - Limited dynamic range (depends on materials)
 - Accuracy depends on unpredictable drift and noise
 - Not flexible for any design change

- Benefits of digital signal processing
 - Flexibility
 - Programmability
 - Multiplexing
 - Predictable and accurate performance
 - Design tools are well developed



Typical DSP system



DSP basic operations(1)

- ◇ Frequency selective and adaptive filtering :
 - Enhances signal-to-noise by selecting frequency bands to pass
 - FIR, IIR, ...
- ◇ Time-frequency transformation :
 - Finds frequency contents
 - FFT, DCT, ...
- ◇ Sample rate change :
 - Changes sampling rate
 - Interpolation, Decimation, ...
- Correlation :
 - Compares, suppressing random events and amplifying repeated ones



DSP basic operations (2)

Operation	Computation
1. Filtering FIR IIR All Pole General	$y(n) = \sum_{k=0}^{M-1} a_k x(n-k)$ $y(n) = \sum_{k=0}^{N-1} b_k y(n-k) + x(n)$ $y(n) = \sum_{k=1}^{N-1} b_k y(n-k) + \sum_{k=0}^{M-1} a_k x(n-k)$
2. Spectrum Analysis DFT FFT Butterfly $X_0 + jY_0 \rightarrow \boxed{} \rightarrow X_0' + jY_0'$ $X_1 + jY_1 \rightarrow \boxed{} \rightarrow X_1' + jY_1'$	$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/N}$ $X_0' = X_0 + R, \quad Y_0' = Y_0 + I$ $X_1' = X_0 - R, \quad Y_0' = Y_0 - I$ $R = X_1 C - Y_1 S, \quad I = X_1 S + Y_1 C$



DSP basic operations (3)

Operation	Computation
3. Correlation	
Auto	$C_{xx}(m) = \frac{1}{N} \sum_{n=0}^{N-1-m} x(n) x(n+m)$
Cross	$C_{xy}(m) = \frac{1}{N} \sum_{n=0}^{N-1-m} x(n) y(n+m)$
Power Spectrum	$P_{xx} = \text{DFT}(C_{xx})$
Cross-Spectrum	$P_{xy} = \text{DFT}(C_{xy})$
4. Decimation = downsampling	$x_D(n) = \begin{cases} x(n/M), & n/M = \text{integer} \\ 0 & \text{otherwise} \end{cases}$ $X_D(f) = X(Mf)$
Interpolation = upsampling	$x_I(nM) = \begin{cases} x(n), & nM = M, 2M, 3M, \dots \\ \text{Interpol}[x(n), x(n+1)], & \text{between } nM \text{ and } (n+1)M \end{cases}$ $X_I(f) = X(f/M)$



DSP algorithms & applications

DSP Algorithms	System Applications
Speech coding And decoding	Digital cellular phones, personal communication systems, digital cordless phones, multimedia computers, secure communications
Speech encryption And decryption	Digital cellular phones, personal communication systems, digital cordless phones, secure communications
Speech recognition	Advanced user interfaces, multimedia workstations, robotics and Automotive applications, digital cellular phones, personal communication systems, digital cordless phones
Speech synthesis	Multimedia PCs, advanced user interfaces, robotics
Modem algorithms	Digital cellular phones, personal communication systems, digital cordless phones, digital audio broadcast, multimedia computers, wireless computing, navigation, data/facsimile modems, secure communications



DSP algorithms & applications

DSP Algorithms	System Applications
Noise cancellation	Professional audio, advanced vehicular audio
Audio equalization	Consumer audio, advanced vehicular audio
Image compression and decompression	Digital cameras, digital video, multimedia computers, consumer video
Beamforming	Navigation, radar/sonar, signals intelligence
Echo cancellation	Speakerphones, modems, telephone switches



Examples of DSP applications

- ◇ Digital cellular phones
- ◇ Automated inspection
- ◇ Vehicle collision avoidance
- ◇ Voice over Internet
- ◇ Motor control
- ◇ Consumer audio
- ◇ Voice mail
- ◇ Navigation equipment
- ◇ Audio production
- ◇ Videoconferencing
- ◇ Pagers
- ◇ Music synthesis, effects
- ◇ Satellite communications
- ◇ Seismic analysis
- ◇ Secure communications
- ◇ Tapeless answering machines
- ◇ Sonar
- ◇ Cordless phones
- ◇ Digital cameras
- ◇ Modems (POTS, ISDN, cable,...)
- ◇ Noise cancellation
- ◇ Medical ultrasound
- ◇ Patient monitoring
- ◇ Radar



Most DSP tasks

- ◇ Most DSP tasks(algorithms) require
 - Repetitive numeric computations
 - Attention to numeric fidelity
 - High memory bandwidth, mostly via array accesses
 - Real-time processing

- ◇ Processors must perform these tasks efficiently while minimizing:
 - Cost
 - Power
 - Memory use
 - Development time



Implementation Methods of DSP algorithms

- ◇ (S/W on Standard microprocessor)
- ◇ Standard Digital Signal Processor (DSP)
 - Use general purpose commercial DSP chip
 - Ex. TI TMS320CXX, Motorola DSP 56000, ...
- ◇ Application-Specific DSP
 - Includes an optimized datapath for special application
 - Ex. AS-DSP for HDTV or Cellular communications
- ◇ Algorithm-Specific DSP (or H/W engine)
 - Includes algorithm-specialized DSP core or H/W engine
 - Ex. Digital filter, FFT...



DSP Implementation Issues

◇ Tradeoffs

- Hardware vs software (programmable DSP)
- Pipelining vs parallelism

◇ Finite word-length arithmetic

- Quantization error, Rounding, overflow, underflow...
- Fixed-point vs floating-point



Standard DSP vs. AS-DSP

Standard DSP	Application-Specific DSP
◆ General purpose for any DSP algorithms ◆ Standardized hardware structure ◆ Emphasis on short design time ◆ Highly flexible ◆ Low cost due to wide applicability of H/W ◆ Low performance/throughput ◆ High power consumption ◆ Not cost effective	◆ Only for a limited set of DSP algorithms ◆ Optimized Datapath & on-chip memory ◆ Higher performance/throughput ◆ Low power consumption ◆ Need large design effort ◆ Inflexibility ◆ Higher initial cost

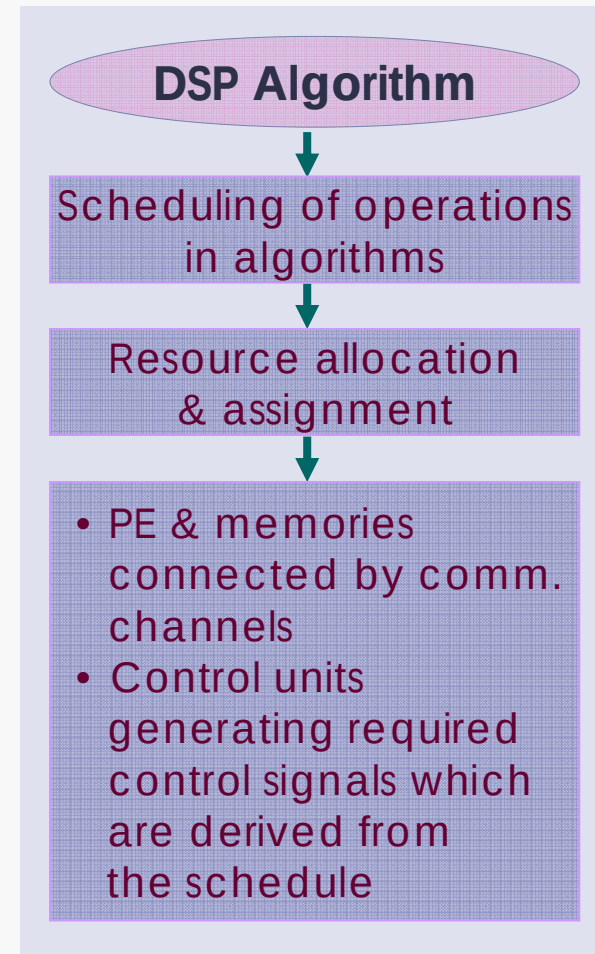


Algorithm-Specific DSP

- ◇ Suitable for implementing Systems with a fixed function (ex. digital filters)

- ◇ Use Direct mapping techniques

- Three design steps 
- Algorithms-Specialized DSP cores



Modeling of DSP Systems

◇ Behavioral description

- I/O description that defines the required action of a system in response to prescribed inputs

◇ Functional description

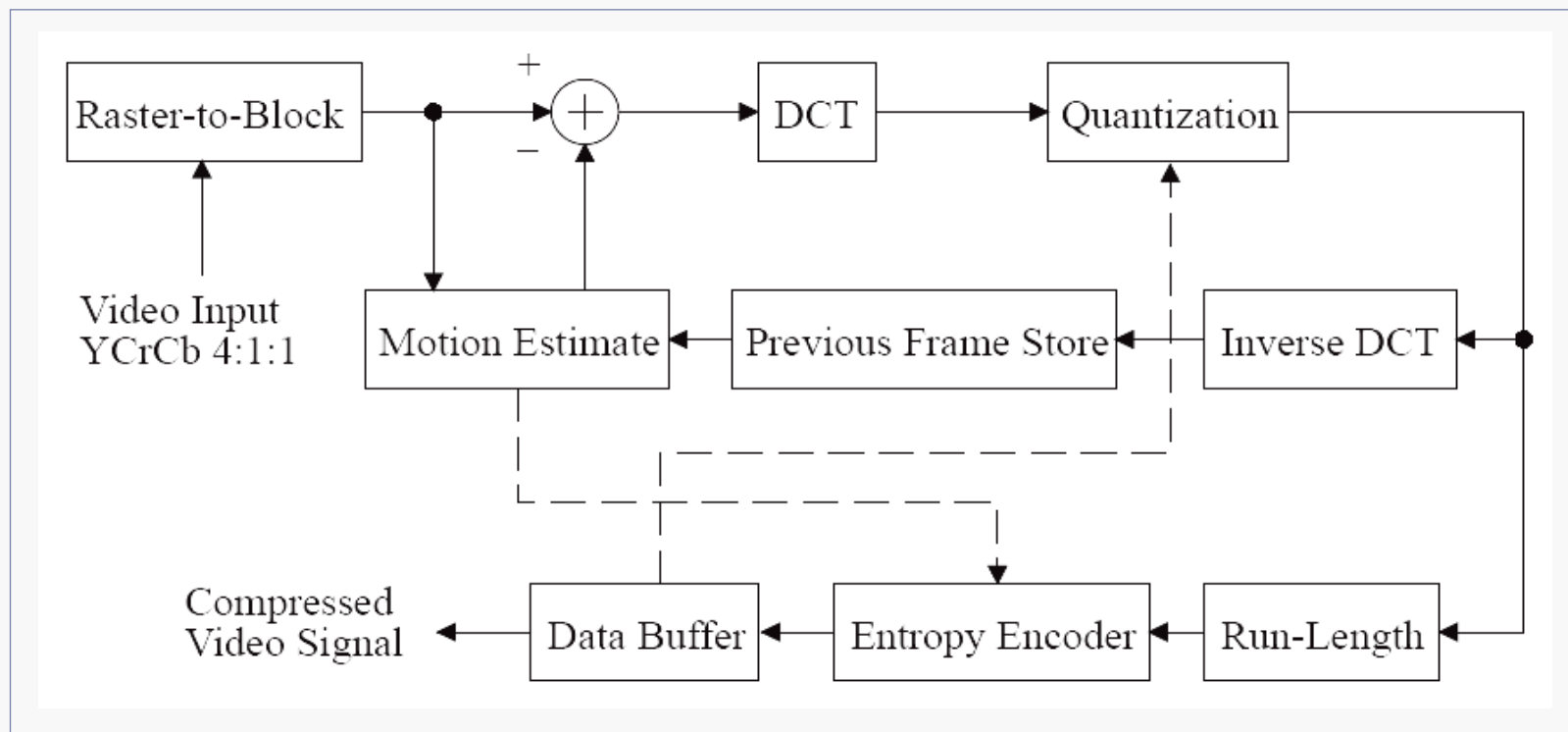
- Defines the manner in which the system is operated to perform its function (using a data-flow model)
- May contain information such as requirements and fractional descriptions of the blocks

◇ Architectural (Structural) description

- Describes how a number of components are interconnected
- May include information about estimated performance



Functional View of H.261 Video encoder



DSP System Design

◇ Selection of Design Methodology

- Overall strategy to organize and solve the design tasks at the different steps of the design process
- Design process is viewed as the development of a sequence of models of the system, where each model is more refined than the previous

◇ Structured design methods are used to

- Guarantee that the performance goals are met (Accuracy, Speed, Power, Chip size, # of pins, ...)
- Attain a short and predictable design time



System Design Phases

System Specification



System Design



VLSI Design



**Manufacturing
& Testing**

- System Description (C/HDL)
- System Partitioning
- RTL & Logic Design



System Specification

◇ System Specification

- A description of the tasks that the system shall perform
- Include requirements on size, power, life cycle, costs, etc
- A time schedule for design completion, production, and market introduction

◇ Two main parts of system Spec.

- 1) Behavioral description that specifies what is to be designed
- 2) Verification or validation that describes how the design should be verified (validated) → Test Plan
 - **Verification**
 - Involves a formal process of proving the equivalence of two different types of representations under all specified conditions
 - **Validation**
 - An informal and less rigorous correctness check
 - Usually done by simulating the circuit with a finite set of input stimuli to assert that the circuit operate correctly



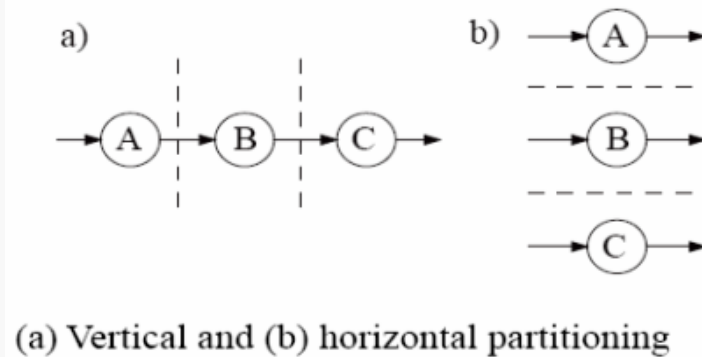
System Design Phase

- ◇ System design phase consists of a sequence of partitioning steps
- ◇ System partitioning techniques
 - 1) Data-Flow approach**
 - Partition the system along the data-flow in the system
 - Vertical or horizontal partitioning
 - 2) Top-Down approach**
 - System is successively partitioned into a hierarchy of subsystems
 - Behavioral description → Structural → Functional
 - 3) Bottom-Up approach**
 - Starts by successively assembling well-known building blocks into more complex blocks until the whole system is realized
 - 4) Meet-in-the-middle approach**
 - Specification-synthesis process is carried out in top-town fashion, but actual design of building blocks is performed in bottom-up fashion

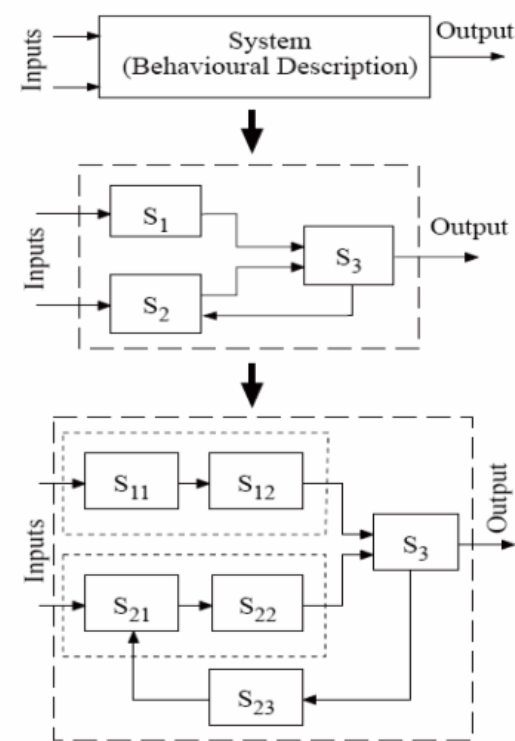


Three Partitioning approaches

Data-Flow Approach



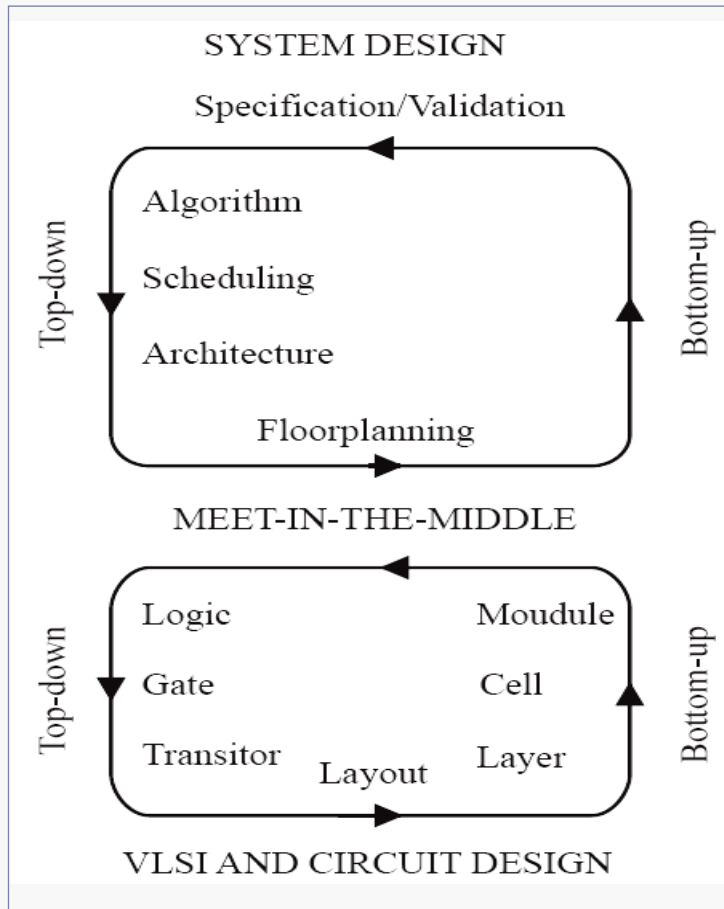
Top-Down Approach



Bottom-Up Approach

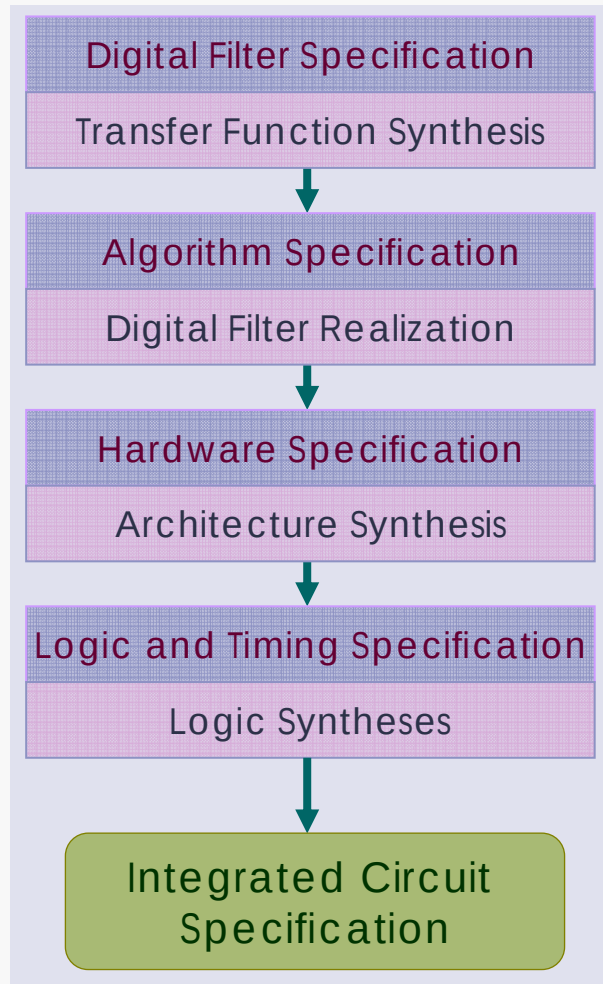
- ◇ Top-down design strategy is generally used for system design

Meet-In-The-Middle Approach



- Design process is divided into two almost independent parts that meet in the middle

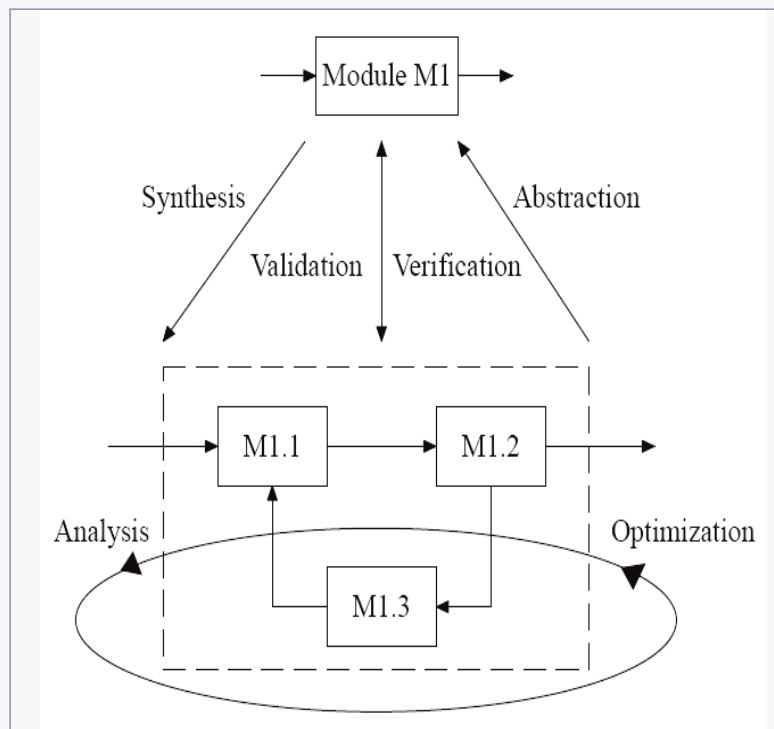
Example of Digital Filter Design



- ◇ Filter spec.
 - Passband, stopband, & sampling freq. and corresponding attenuations
 - A transfer function meeting the spec. is determined
- ◇ Algorithm spec.
 - A suitable algorithm to realize the filter is proposed (sampling rate, dynamic signal range, ...)
- ◇ H/W spec.
 - Arithmetic operations in algorithm are scheduled
 - Mapping the operations onto a suitable S/W & H/W architecture
- ◇ Logic design
 - Logic design of architectural components (PE, memories, channels, & CU)



Design Transformations



- ◇ Synthesis : (adds details)
 - One level → lower level
- ◇ Abstraction : (Hides details)
 - One level → higher level
- ◇ Optimization
 - Same abstraction level
- ◇ Analysis
 - Supports decision making
- ◇ Verification & Validation
 - Assure the representations correctness between two levels

VLSI Design phase

◇ VLSI design phase

- A process of transforming the structural description of logic circuitry into a detailed physical layout
- *Turnaround time* for design changes
 - Ranges from several weeks to many months
 - Long design times lead to last opportunities of marketing the chip ahead of the competition and recouping the investment
- *Correctness of the design*
 - Of paramount importance for a successful project
- Design process involves technical trade-offs between system performance, costs, & design efforts
- Technical feasibility study of the whole system is an essential step



Technical Feasibility Study (1)

◇ System-related

- Partitioning into cabinets, boards, and circuits
- Mixed digital and analog circuits on the same chip
- Clock frequencies
- Power dissipation and cooling
- Circuit area and packaging
- I/O interface

◇ Circuit-related

- External
 - Interchip propagation delay
 - Data transfer frequencies
 - Input protection
 - Loads that have to be driven, including expected PCB runs
 - Available input drivers
 - Drive capacity for output buffers
 - Restrictions on pin-outs



Technical Feasibility Study (2)

◇ Circuit-related

■ Internal

- Clock frequencies
- Data transfer frequencies and distances
- Critical timing paths
- Non critical timing paths
- Power dissipation and cooling
- Drive capacity for internal buffers
- Circuit area, yield, and packaging
- Temperature and voltage effects
- Maximum and minimum temperatures, voltages, etc
- Process technology

◇ Design-effort-related

- CAD tools
- Layout style
- Regularity and modularity of the circuits
- Module generators
- Cell library



Next Topics to cover

- ◇ Review of DSP theory
- ◇ DFT & FFT algorithm and architecture
- ◇ Digital filter design
- ◇ DSP VLSI architecture
- ◇ DSP hardware elements
- ◇ S/W tools for DSP system
- ◇ DSP applications

